Case 03 · Spec on one page: Roll Transparency Log

A summary of the full specification: the promise, the ten features, the key decisions, and what stays open.

What the product promises

- Every change to the roll, with its reason, is kept so that it cannot be altered quietly.
- Every person checks their own line in a second, and receives a proof.
- No names on the log, and no need to trust whoever runs it.

Ten features

#	Feature	Layer	In one sentence
1	Change capture	Intake	Every officer action reaches the log before it takes effect
2	Reason codes	Intake	11 codes with evidence; no "other"
3	Keyed identifiers	Intake	HMAC pseudonyms; the key stays in an HSM
4	Merkle log	Log	Append-only; a rewrite is detectable
5	Signed tree heads	Log	Every minute, cosigned by 3 of 5 witnesses
6	Reconciliation	Log	The published roll must equal the log, booth by booth
7	Self-check	Voter	Status, history, reason and appeal deadline, with proof
8	Alert and appeal	Voter	SMS on the day of change; pre-filled appeal
9	Witness network	Witnesses	Independent bodies check the log only grows
10	Aggregate data	Witnesses	District and week only; cells under 10 suppressed

Key decisions

- The Commission runs the log; witnesses approve it.
- Identifiers are keyed in an HSM.
- A change is logged before it takes effect.
- The public sees aggregates; the person sees their own line.

What stays open

- The log proves what was recorded, not that it was right.
- It shortens the time until a person knows, not the appeal queue.